

The empirical recurrence for OEIS A237137: recovering a conjecture that is not written down

Adrian Perez Fontelles
Independent researcher

2 September 2026

Abstract

OEIS A237137 records “Empirical recurrence of order 93”, with the recurrence itself in a linked file rather than in the entry. The conjecture can nevertheless be settled, and the recurrence recovered, without reading that file. The entry counts arrays under a condition local to a bounded window of lines, so the count is a walk count on a finite digraph and satisfies some monic recurrence of order at most the number of its states with distinct futures, here $S' = 714$; Berlekamp–Massey applied to $2S'$ exact terms therefore returns the MINIMAL such recurrence. Its order is exactly 93, the order the entry states, and the same is true of every tail of the sequence. Any recurrence of order 93 that the sequence satisfies from any point on must then have a characteristic polynomial that is a multiple of the minimal one and of the same degree, hence equal to it. So the entry’s recurrence is the one displayed here, whatever its file contains, and it is proved.

2020 Mathematics Subject Classification. 05A15, 11B37, 15A18.

1 A conjecture one cannot read

OEIS A237137 is “Number of $(n+1) \times (3+1)$ 0..3 arrays with the maximum plus the upper median of every 2×2 subblock equal.”. It has offset 1 and begins

8414, 333446, 9999802, 436271890, 17759803582, 786068103136, 34365943860434, 1528315973150030, ...

The entry states:

Empirical recurrence of order 93 (see link above)

As of the “Last modified” line on the live entry (Jul 23 2025 09:52:24, revision 7) this is still recorded as empirical. The recurrence’s coefficients are not in the entry text.

2 The count is a walk count

The entry’s condition is local: it constrains a bounded window of consecutive lines of the array and nothing further. The admissible configurations of one such window are the vertices of a finite digraph, an edge joining u to v when v may follow u , and an admissible array is exactly a walk. Hence, with M the adjacency matrix on $S = 714$ vertices, $a(n) = \iota^\top M^{j(n)} \tau$ for a linear reindexing j fixed by the entry’s shape and checked against its published terms. By the Cayley–Hamilton theorem M^S is an integer combination of $I, M, \dots, M^{S-1}$, so:

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 714$.*

3 Recovering the recurrence

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S is determined, together with its minimal recurrence, by its first $2S$ terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Running it on $2S' = 1428$ exact terms of the model returns order 93 — the order the entry states — with the integer coefficients

$$a(n) = \sum_{i=1}^{93} c_i a(n-i),$$

c_1	84	c_{25}	−540517308142144444206622445976	c_{49}	125561
c_2	−333	c_{26}	413375477679417581330967750196	c_{50}	105494
c_3	−135993	c_{27}	23077486149928249494753648015076	c_{51}	−11579
c_4	2556265	c_{28}	−40522812642213407116927015302615	c_{52}	−58176
c_5	87744206	c_{29}	−801143945485364244715422910643066	c_{53}	815555
c_6	−2548246972	c_{30}	2098297710336186225140343572091922	c_{54}	269375
c_7	−27322507697	c_{31}	22708848992443008166707093542421800	c_{55}	−461390
c_8	1290703562706	c_{32}	−77977872861660833073440069385350060	c_{56}	−104211
c_9	2796551575389	c_{33}	−526190635147614546172617469554083837	c_{57}	2143553
c_{10}	−408563770238163	c_{34}	2251245786262843315560175785789760437	c_{58}	3339103
c_{11}	982785315189292	c_{35}	9940148039957615545535291592640053491	c_{59}	−826846
c_{12}	88321963656309703	c_{36}	−52233767995480709902718672736670962101	c_{60}	−872523
c_{13}	−485336802202616717	c_{37}	−151749343976031181007561583071253479374	c_{61}	2662549
c_{14}	−13674495788032242731	c_{38}	991944937654619211717837766946161402040	c_{62}	1804026
c_{15}	110179621808540144532	c_{39}	1833156607619964631726641074145025839346	c_{63}	−7172119
c_{16}	1554786004564628316515	c_{40}	−15591526135477735790408442626342347067730	c_{64}	−2747822
c_{17}	−16595137187141909500092	c_{41}	−16617399932147023499086530284765623529608	c_{65}	16155326
c_{18}	−130637631833176815037789	c_{42}	204336824911439668111809278241622268659260	c_{66}	2365843
c_{19}	1828912622344955231584338	c_{43}	94224486376327353296584622926043108085000	c_{67}	−3035616
c_{20}	7945229116119473549586416	c_{44}	−2243934172393096359778425268937152136584432	c_{68}	1497279
c_{21}	−154267929554827564418080449	c_{45}	54859786385041907249624322185076306708944	c_{69}	47369607
c_{22}	−318723808355525349341362011	c_{46}	20714765960375492681595285876431512313564176	c_{70}	−1023890
c_{23}	10219518264503531032631598671	c_{47}	−8887520596713128353647767951387144114611952	c_{71}	−6097338
c_{24}	4666978591580774196937011094	c_{48}	−161044825747381851109183071019957748140962288	c_{72}	22157432

Theorem 3. *The recurrence above holds for every n , and it is the recurrence the entry records.*

Proof. It holds by Lemma 2, which returns a recurrence the sequence satisfies from its first term. For the identification: the entry asserts that the sequence satisfies SOME recurrence of order 93, possibly only from some index on. Let q be that recurrence’s characteristic polynomial and $q_{\min}$ the minimal polynomial of the tail of the sequence. Then $q_{\min} \mid q$. Berlekamp–Massey run on the sequence with its first $93 + 4$ terms removed returns order 93 as well, so $\deg q_{\min} = 93 = \deg q$; both are monic, so $q = q_{\min}$, which is the polynomial computed above. $\square$

4 Verification

Three checks.

First, the walk model reproduces all 13 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters.

Second, the recovered recurrence was evaluated directly on those published terms, with no matrices and no Berlekamp–Massey involved, and holds at every index where the entry gives enough earlier terms to test it.

Third, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once in exact rational arithmetic; and once more on the sequence with its first $93 + 4$ terms removed, which is what the identification above requires. All three agree.

Remark 4. What is unusual here is that the conjecture’s statement was never read. The entry pins it down to a one-parameter family — the recurrences of order 93 that this sequence satisfies — and that family turns out to have exactly one member.

References

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